

University of Nevada – Reno
Computer Science & Engineering Department
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Network Security

Guest Lecture

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Network security

❖ field of network security:

- how bad guys can attack computer networks
- how we can defend networks against attacks
- how to design architectures that are immune to attacks

❖ Internet not originally designed with (much) security in mind

- *original vision*: “a group of mutually trusting users attached to a transparent network” 😊
- Internet protocol designers playing “catch-up”
- security vulnerabilities in all layers!

Security: Definition

- ❖ Security is a state of well-being of information and infrastructures in which the possibility of theft, tampering, and disruption of information and services is kept low or tolerable

The Cast of Characters

- ❖ Alice and Bob are the good guys



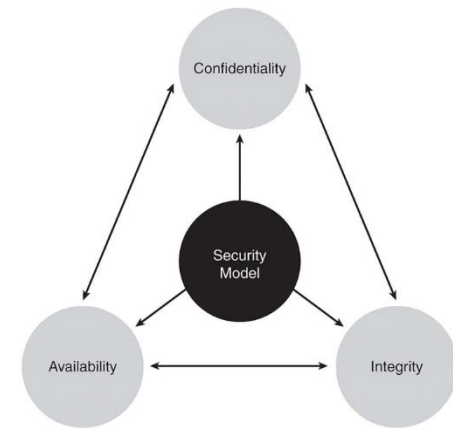
- ❖ Trudy/Mallory is the bad guy
- ❖ Trudy is our generic “intruder”



- ❖ Who might Alice, Bob be?
 - ... well,
 - Web browser/server for electronic transactions
 - on-line banking client/server
 - DNS servers
 - routers exchanging routing table updates

Alice's Online Bank

- ❖ Alice opens Alice's Online Bank (AOB)
- ❖ AOB must prevent Trudy from learning Bob's balance
 - **Confidentiality** (prevent unauthorized reading of information)
- ❖ Trudy must not be able to change Bob's balance
- ❖ Bob must not be able to improperly change his own account balance
 - **Integrity** (prevent unauthorized writing of information)
- ❖ AOB's information must be available when needed
 - **Availability** (data is available in a timely manner when needed)



Alice's Online Bank

- ❖ How does Bob's computer know that "Bob" is really Bob and not Trudy?
- ❖ When Bob logs into AOB, how does AOB know that "Bob" is really Bob?
 - **Authentication** (assurance that other party is the claimed one)
- ❖ Bob can't view someone else's account info
- ❖ Bob can't install new software, etc.
 - **Authorization** (allowing access only to permitted resources)
- ❖ Bob can't deny a transaction he requested
 - **Non-repudiation** (protection against denial by one of the parties in a communication)

Terminology: Security Threats and Attacks

- ❑ A **threat** is a potential violation of security
 - Flaws in design, implementation, and operation
- ❑ An **attack** is any action that violates security
 - Active vs. passive attacks
- ❖ **Security Services**
 - Enhance the security of data processing systems and information transfers of an organization
 - Designed to prevent, detect or recover from a security attack.
 - Firewall, IDS, Digital Forensics

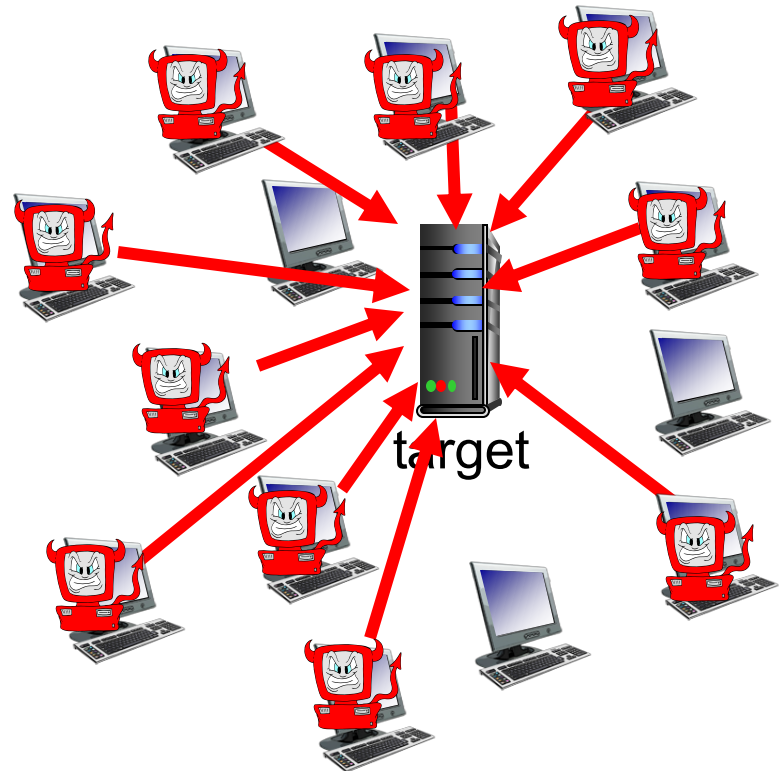
What can the attackers do:

- ❖ Put malware into hosts via Internet
- ❖ malware can get in host from:
 - *virus*: self-replicating infection by receiving & executing object (e.g., e-mail attachment)
 - *worm*: self-replicating infection by passively receiving object that gets itself executed
- ❖ *spyware malware* can record keystrokes, websites visited, upload info to collection site
- ❖ infected host can be enrolled in *botnet*, used for spam, DDoS attacks

What can the attackers do:

Denial of Service (DoS): attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with bogus traffic

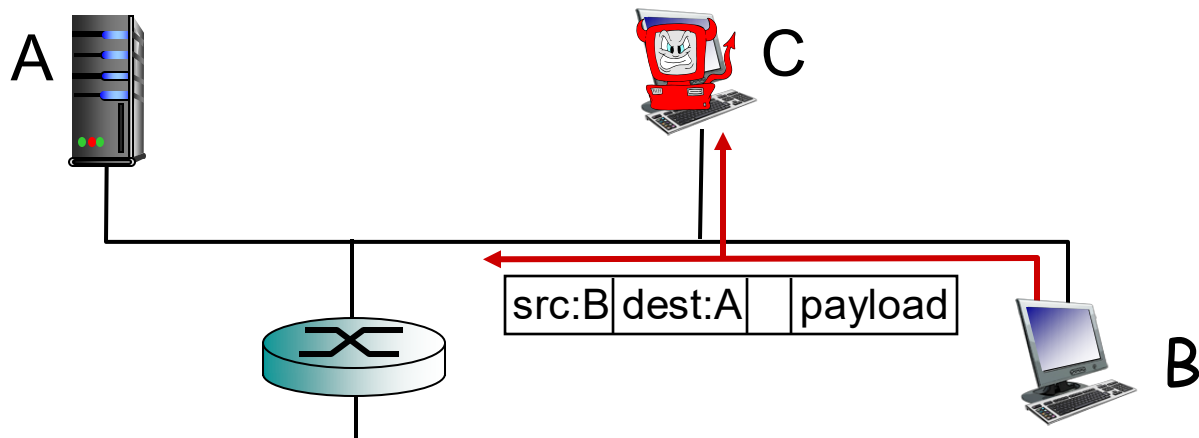
1. select target
2. break into hosts around the network (see botnet)
3. send packets to target from compromised hosts



Attacker can sniff packets

packet “sniffing”:

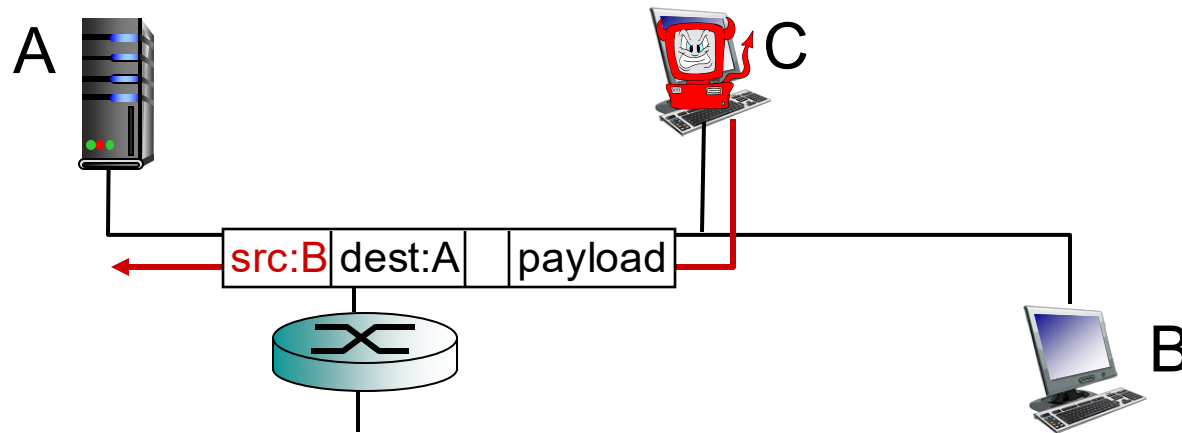
- broadcast media (shared ethernet, wireless)
- promiscuous network interface reads/records all packets (e.g., including passwords!) passing by



- ❖ wireshark software used for labs is a (free) packet-sniffer

Attacker can use fake addresses

IP spoofing: send packet with false source address



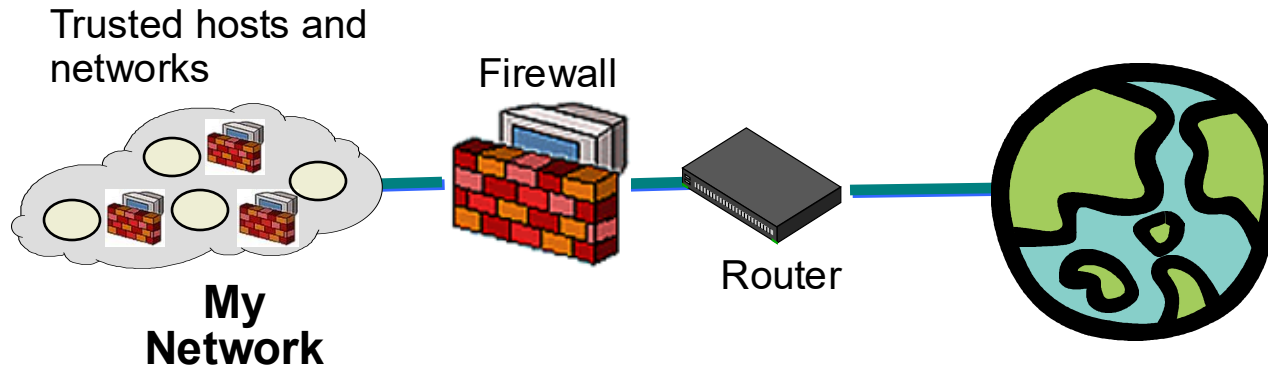
Defending Against Network Security Attacks

- ❖ Well, those all sound pretty terrible!!
- ❖ So what do I do to keep my networks safe from security attacks?
- ❖ How would I even go about starting to defend myself and others from variations of attacks?

Some network security
mechanisms terminology

Firewalls

- Idea: separate local network from the Internet



Firewall Characteristics

Four general techniques mostly used

- ❑ **Service Control:** Determines the types of Internet services that can be accessed
- ❑ **Direction control:** Determines the direction in which particular service requests may be initiated and allowed to flow through the firewall
 - inbound or outbound

Firewall Characteristics (contd.)

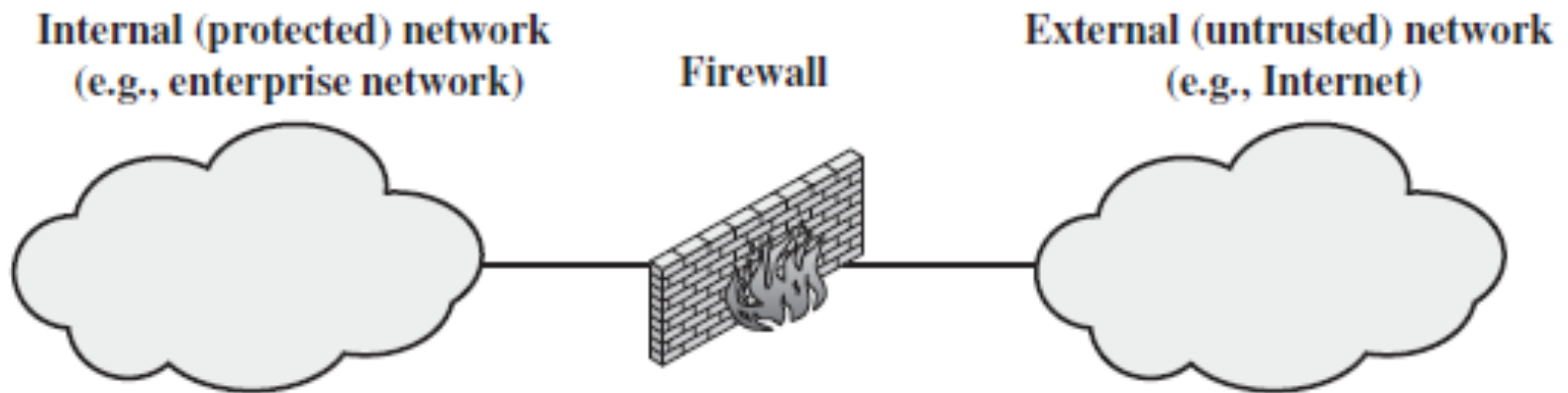
- ❖ **User control:** Controls access to a service according to which user is attempting to access it
 - Typically applied to users inside the firewall perimeter
- ❖ **Behavior control:** Controls how particular services are used

Firewall Locations

- ❖ Where to filter/ place firewall?
- ❖ Balance between risk and costs
- ❖ While placing firewall, remember:
- ❖ Always a higher layer that is hard to filter
 - Humans

Firewall Locations (2)

- ❖ Typically a firewall is positioned to provide a protective barrier between an external, potentially untrusted source of traffic and an internal network



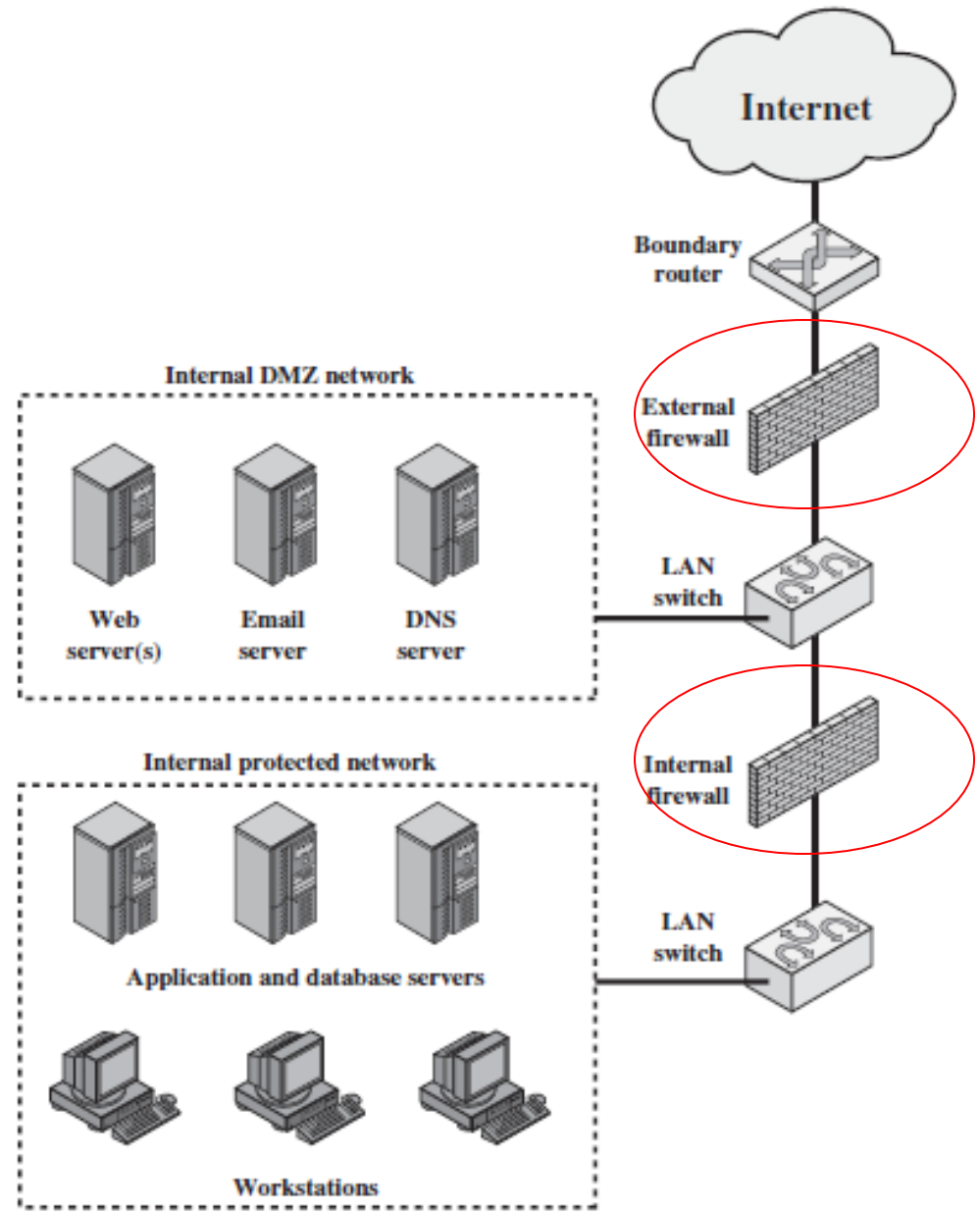
Firewall Popular Locations

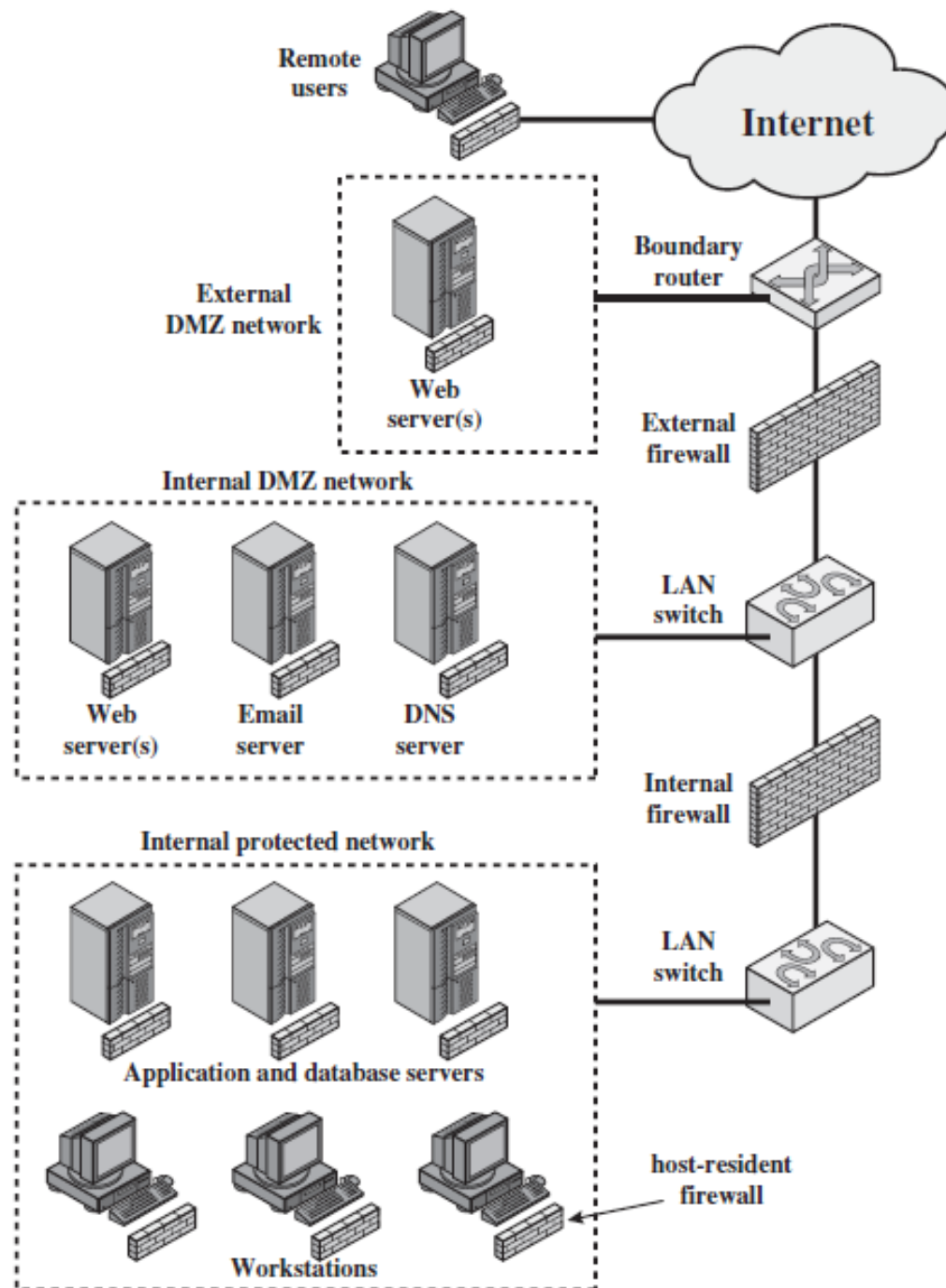
- ❖ Between internal LAN and external network
- ❖ At the gateways of sensitive subnetworks within the organizational LAN
 - Payroll's network must be protected separately within the corporate network
- ❖ On end-user machines
 - “Personal firewall”



Firewall Locations – intro of DMZ

- ❖ DMZ (demilitarized zone) network

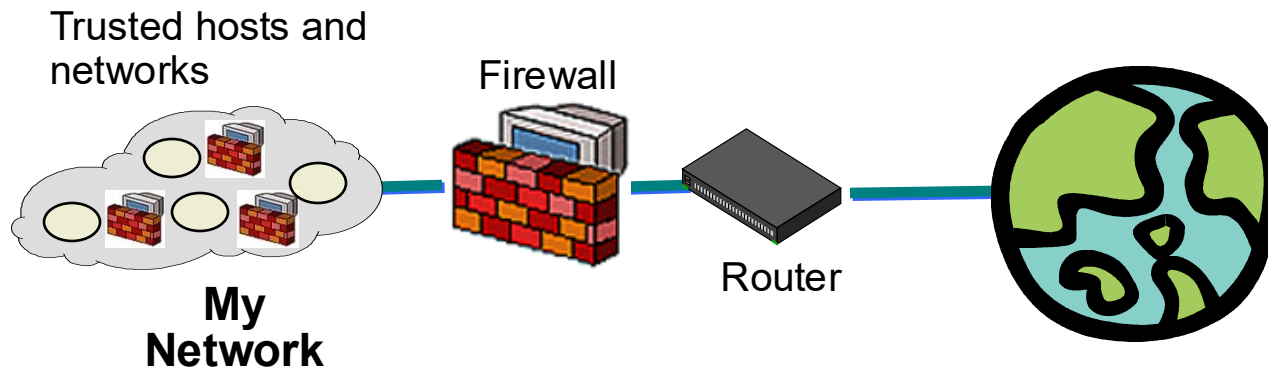




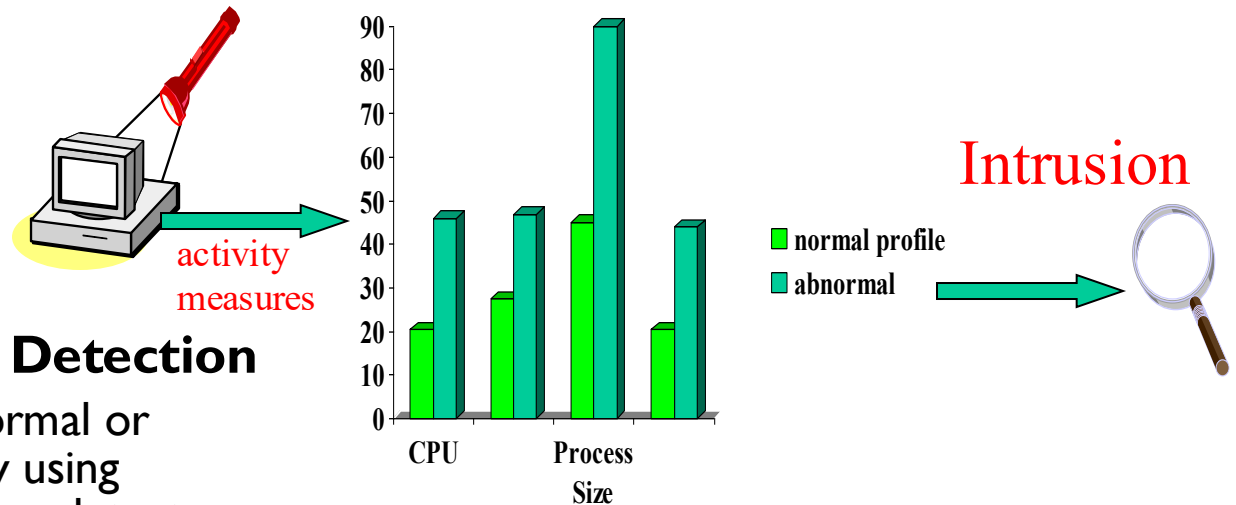
Distributed
Firewall
Configuration

Intrusion Detection System (IDS)

- ❖ The implementation of IDS and its correctness is important
 - Equally important is its placement in the network topology
- ❖ Depends on what type of activities you want to detect
 - Internal,
 - External or
 - Both



Intrusion Detection Approaches

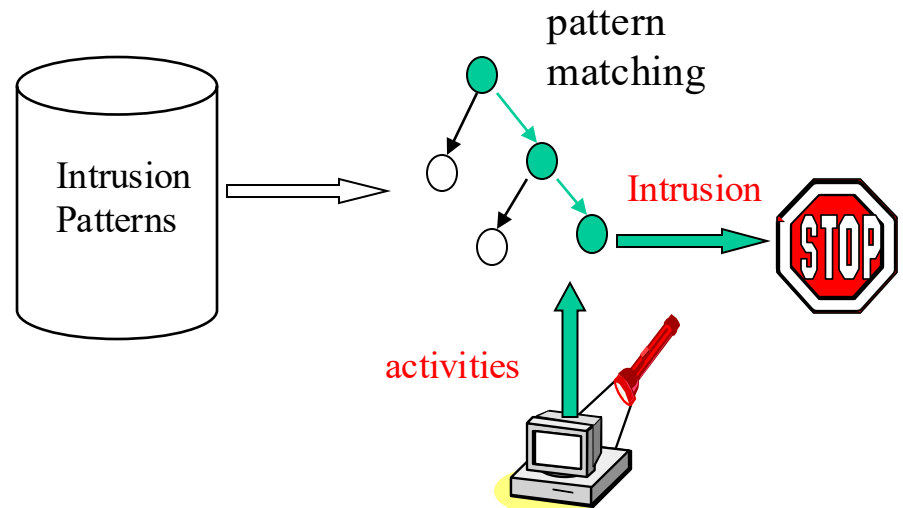


□ Statistical Anomaly Detection

- Attempt to define normal or expected behavior by using statistical data and then detect intrusion

□ Rule-Based Detection

- Attempt to define proper behavior by using a set of rules and then detect intrusion



Summary

- ❖ What is the field of network security
- ❖ How attackers attack
- ❖ Quick overview into defense mechanisms
- ❖ History: Reading for home

Reading for home
History: Did you know?

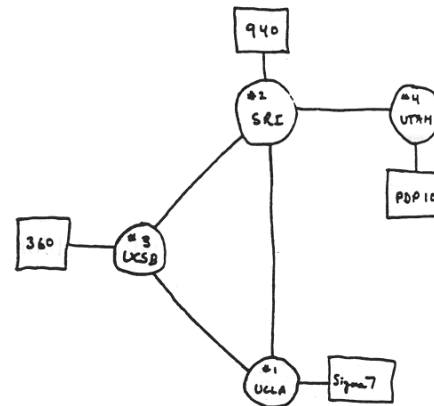
Internet history

1961-1972: Early packet-switching principles

- ❖ **1961:** Kleinrock - queueing theory shows effectiveness of packet-switching
- ❖ **1964:** Baran - packet-switching in military nets
- ❖ **1967:** ARPAnet conceived by Advanced Research Projects Agency
- ❖ **1969:** first **ARPAnet** node operational

❖ **1972:**

- ARPAnet public demo
- NCP (Network Control Protocol) first host-host protocol
- first e-mail program
- ARPAnet has 15 nodes



THE ARPA NETWORK

Internet history

1972-1980: Internetworking, new and proprietary nets

- ❖ **1970: ALOHAnet** satellite network in Hawaii
- ❖ **1974:** Cerf and Kahn - architecture for interconnecting networks
- ❖ **1976:** Ethernet at Xerox PARC
- ❖ **late 70' s:** proprietary architectures: DECnet, SNA, XNA
- ❖ **late 70' s:** switching fixed length packets (ATM precursor)
- ❖ **1979:** ARPAnet has 200 nodes

Cerf and Kahn' s internetworking principles:

- minimalism, autonomy - no internal changes required to interconnect networks
- best effort service model
- stateless routers
- decentralized control

**define today' s Internet
architecture**

Internet history

1980-1990: new protocols, a proliferation of networks

- ❖ **1982:** smtp e-mail protocol defined
- ❖ **1983:** deployment of **TCP/IP**
- ❖ **1983:** DNS defined for name-to-IP-address translation
- ❖ **1985:** ftp protocol defined
- ❖ **1988:** TCP congestion control
- ❖ new national networks: Csnet, BITnet, NSFnet, Minitel
- ❖ 100,000 hosts connected to confederation of networks

Internet history

1990, 2000 's: commercialization, the Web, new apps

- ❖ early 1990' s: ARPAnet decommissioned
- ❖ 1991: NSF lifts restrictions on commercial use of NSFnet (decommissioned, 1995)
- ❖ early 1990s: Web
 - hypertext [Bush, Nelson]
 - HTML, HTTP [Berners-Lee]
 - 1994: **Mosaic**, later Netscape
 - late 1990' s: commercialization of the Web
- late 1990' s – 2000' s:
 - ❖ more killer apps: instant messaging, P2P file sharing
 - ❖ network security to forefront
 - ❖ est. 50 million host, 100 million+ users
 - ❖ backbone links running at Gbps

Internet history

2005-present

- ❖ ~750 million hosts
 - Smartphones and tablets
- ❖ Aggressive deployment of broadband access
- ❖ Increasing ubiquity of high-speed wireless access
- ❖ Emergence of online social networks:
 - Facebook: one billion users
- ❖ Service providers (Google, Microsoft) create their own networks
 - Bypass Internet, providing “instantaneous” access to search, email, etc.
- ❖ E-commerce, universities, enterprises running their services in “cloud” (eg, Amazon EC2)

Early Hacking – Phreaking

- ❖ In 1957, a blind seven-year old, Joe Engressia Joybubbles, discovered a whistling tone that resets trunk lines
 - Blow into receiver – free phone calls



- Cap'n Crunch cereal prize
 - Giveaway whistle produces 2600 MHz tone



The Eighties

❖ Robert Morris worm - 1988

- Developed to measure the size of the Internet
 - However, a computer could be infected multiple times
- Brought down a large fraction of the Internet
 - ~ 6K computers



- Academic interest in network security

The Nineties

❖ Kevin Mitnick

- First hacker on FBI's Most Wanted list
- Hacked into many networks
 - including FBI
- Stole intellectual property
 - including 20K credit card numbers
- In 1995, caught 2nd time
 - served five years in prison



The Twenties

❖ Code Red worm

- Jul 19, 2001: infected more than 359K computers in less than 14 hours

❖ Sapphire worm

- Jan 31, 2003: infected more than 75K computers (most in 10 minutes)

❖ DoS attack on sco.com

- Dec 11, 2003: SYN flood of 50K packet-per-second

❖ Nyxem/Blackworm virus

- Jan 15, 2006: infected about 1M computers within two weeks